Project #2 - Team Seven Proposal

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Project title: *Coming Clean - What Toronto Airbnb Reviews Reveal*

Kaggle dataset: Dataset link: Toronto Airbnb Dataset

Source: Inside Airbnb: Toronto (Sep 2026)

Number of observations: 729,202 guest review records from June 6th, 2010 - September 17th, 2026. Each row represents an individual guest review of a Toronto Airbnb listing.

Relevant variables: listing_id, ID, date, reviewer_id, reviewer_name, comments

Research question: *What themes appear in Toronto Airbnb guest reviews, and which are most common in reviews expressing dissatisfaction?*

Intended users / decisions:

- **Airbnb:** Based on complaints, which listings are bad? How can we improve listing quality?
- **Customers/Guests**: Based on glowing reviews, where are the good listings?
- **Hosts**: Based on customer sentiment, is my listing good or bad? What should I fix?

Analytical plan:

1. Preprocessing: executive decision: drop all non english reviews, as opposed to translating english reviews
2. Join host_name by listing_id to reviews.csv, possibly listing_ratings by listing_id for a better picture
3. Bag-of-words the reviews
4. Tokenization and determine stop words: replace found host_names with generic Host_Name and anonymize or remove reviewer_name (unnecessary)
5. Flag stop words that flip sentiments with valuable tokens
6. Remove unnecessary stop-words

7. Lemmatization: self explanatory, also account for host_name nicknames
8. Calculate tf, idf, then tf-idf
9. Perform LDA topic modeling (discussed, haven't learned yet)
10. Cross analyze results/outcomes of tf-idf with LDA topic modeling

Performing the primary text-mining technique/s and the simple supporting analysis/es all serve to convert data into information and provide evidence to answer our questions about guest reviews for AirBnBs. The goal is to identify patterns, themes, semantics, and emotions that surround a guest's words in their reviews in order to provide insight to other guests, ideas for improvements to hosts, and genuine concerns to AirBnB. AirBnB's website has simple search features for text in reviews but instead of generalizing and summarizing the reviews, single words get tagged in a filter-based search which provides an opening to analyze the text of the reviews more deeply and in different ways than AirBnB already does.

Scope check: CSV file is easily downloaded, extracted from a zip, and loaded into a dataframe inside of a Google Colaboratory notebook and as the CSV is only 700k rows and only 208 MB in size, will not outstrip computing resources or capabilities.

> **Potential data limitations:** Quality of reviews. Character/Word length of reviews. Amount of reviews per listing. Number of listings reviewed. Language differences. Historical coverage - the data only includes records from 2010-2026, nothing earlier.
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> **Worthwhile topic which will be relevant**: Michelin star question: how relevant is the guest's interaction with the host to the rating of an airbnb listing?